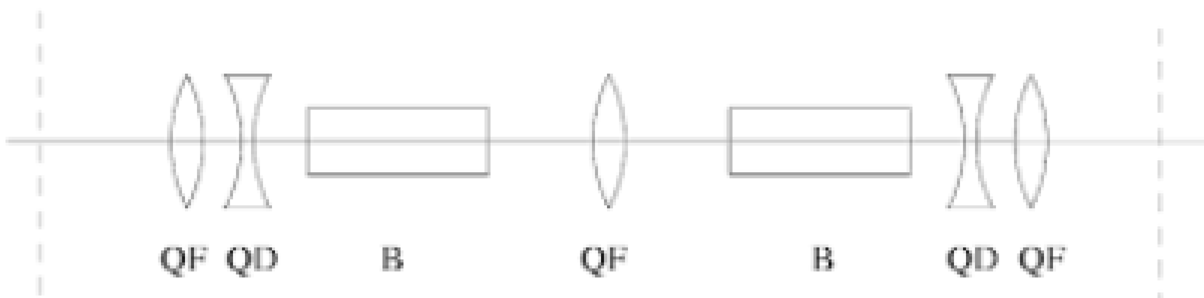
Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. NSLS II Longitudinal Parameters (Electrons) 50 pts.

NSLS II adopts a double bent achromat, DBA, lattice.



Parameters	Values
Energy [GeV]	3.0
Circumference [m]	780
Number of dipoles	60
Dipole field [T]	0.4
Beam current [A]	0.5
RF frequency [MHz]	499.68
Harmonic number	1320

From the design parameters of NSLS II shown in the table above, calculate the following:

a Calculate the axial length of the dipoles assuming all dipoles are the same. (5pts)

b In the DBA lattice, the dispersion, D, and dispersion slope, D', are zero at one end of the dipoles and non-zero at the other end. Find the dispersion inside the dipole magnet (10 pts).

c Calculate the energy loss due to the dipole field (10 pts).

d If the synchronous accelerating phase of the RF cavity is $\pi/6$, what is the minimum RF voltage required? How much power is required? (5pts)

e The actual RF voltage is about 3 MV. Using this value, find the longitudinal tune of NSLC II (5 pts).

f What is the critical radiation frequency of the dipole radiation? (5pts)

g Find the partition number, $\overline{D}$, due to the synchrotron h Find the longitudinal damping rate, α_E, and compare with the period of longitudinal oscillations (5 pts).

SOLUTION: NSLS II Longitudinal Parameters 50 pts.

a) 5 pts: NSLS II has 60 dipoles to form a closed loop, therefore each dipole bends 6 degree, which is $6/180 * \pi = 0.105$ rad. The radius of the dipole can be found as $P = eB\rho$, therefore $\rho = 3GeV/c/0.4m = 25m$. The length of each dipole is $L_D = 2\pi\rho/60 = 2.618\ m$.

b) 10 pts: Since at one end has $D = 0$ and $D' = 0$, we can calculate the dispersion function in the dipole from this end using small angle approximation:

$$\begin{bmatrix} D(s) \\ D'(s) \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & l & \rho\theta^2/2 \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} D(0) = 0 \\ D'(0) = 0 \\ 1 \end{bmatrix}$$

The dispersion function in dipole is $D(s) = s^2/(2\rho)$.

c) 10 pts:

The energy loss due to synchrotron radiation per turn is

$$U_{SR} = C_\gamma E^4 \oint ds/(2\rho^2\pi) = \frac{C_\gamma E^4}{\rho} = 0.287MeV$$

d) 5 pts: These energy must be compensated by the RF cavity

$$\begin{aligned} eV \sin\phi_s &= U_{SR} \\ V &= 0.57MV \end{aligned}$$

therefore the RF cavity must provide at least 0.57 MV to compensate the synchrotron radiation loss in the dipoles.

The power is current times the voltage - 0.5 A $\times$ 0.3 MV. Power is 150 kW.

e) 5 pts: The compaction factor is

$$\begin{aligned} \alpha_c &= \frac{1}{C}\oint \frac{D(s)}{\rho} ds \\ N \times \alpha_c &= \frac{60}{C}\int_0^{L_D} s^2/(2\rho^2) ds \\ &= \frac{10}{C}\frac{L_D^3}{\rho^2} = 3.68 \times 10^{-4} \end{aligned}$$

for N=60 dipoles.

The actual voltage is 3 MV, the phase should be $\arcsin(0.287/3) = \pi - 0.095$, the synchrotron tune is:

$$\nu_s = \sqrt{\frac{h\eta eV \cos\phi_s}{2\pi E}} = 8.8 \times 10^{-3}$$

f) 5 pts: The time to finish one synchrotron oscillation is $2\pi/(\nu_s\omega_0) = 2.94 \times 10^{-4}s$.
The critical frequency of the dipole radiation is given by

$$\omega_c = \frac{3}{2}\frac{c\gamma^3}{\rho} = 3.64 \times 10^{18} Hz$$

g) 5 pts: We then calculate the radiation integrals, taking advantage that $K(s) = 0$ in dipoles:

$$I_2 = \oint 1/\rho^2 ds = 2\pi/\rho = 0.2513 m^{-1}$$

$$I_3 = \oint 1/\rho^3 ds = 2\pi/\rho^2 = 1.0 \times 10^{-2} m^{-2}$$

$$I_4 = \oint D/\rho^3 ds = 60 \int_0^{L_D} s^2/2/\rho^4 ds = 10 L_D^3/\rho^4 = 4.59 \times 10^{-4} m^{-1}$$

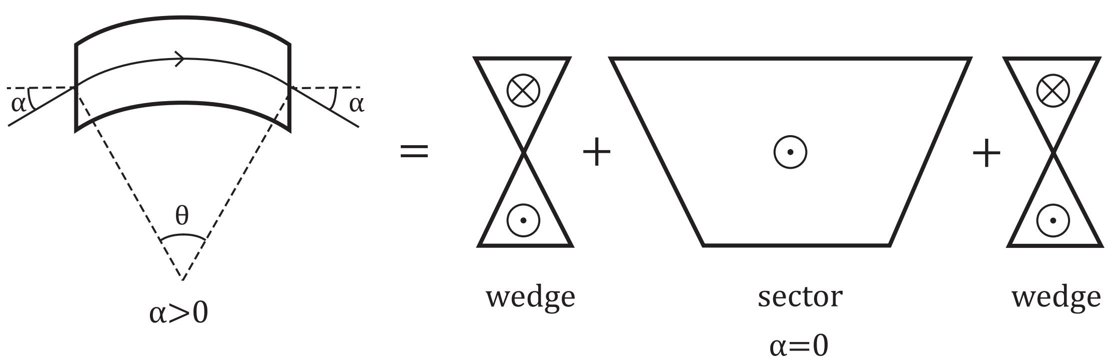
Therefore the partition number $\bar{D} = I_4/I_2 = 1.828 \times 10^{-3}$

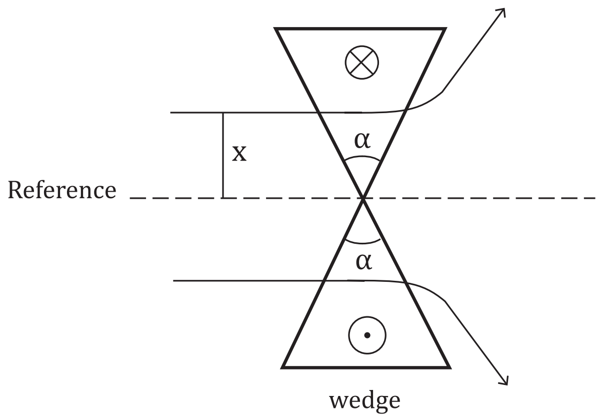
h) 5 pts:
The longitudinal damping rate

$$\alpha_E = \frac{U_0}{2T_0E}(2 + \bar{D}) = 36.79 s^{-1}$$

It is much slower than the synchrotron oscillation.

Problem 2. Dipole Edge Corrections - Horizontal 20 pts.





a) 15 pts: Horizontal correction:

A dipole with entry and exit angles α (defined relative to reference orbit, see figure above) can be divided into three sections: a sector dipole in the center with a wedge magnet on both ends. The sector dipole has the transfer matrix derived in class.

For a short sector, argue from the Lorentz force equation that the particle experiences the following corrections for the orbit displacement x through the wedge:

$$\Delta x' = \frac{B_y(0)}{[B\rho]}\tan(\alpha)x$$
$$\Delta x = 0$$

to show that:

$$\mathbf{M}_{\text{wedge}} = \begin{bmatrix} 1 & 0 \\ \frac{\tan(\alpha)}{\rho} & 1 \end{bmatrix}; \quad \frac{1}{\rho} = \frac{B_y(0)}{[B\rho]}$$

This "kick" correction will be applied entering and exiting the magnet.

b) 5 pts: Show that the x-transfer matrix of the full dipole can be expressed as:

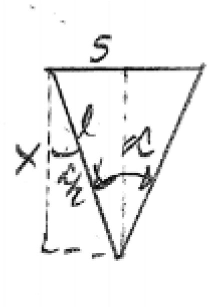
$$\mathbf{M}_x = \mathbf{M}_{\text{wedge}} \cdot \mathbf{M}_{\text{sector}} \cdot \mathbf{M}_{\text{wedge}} = \begin{bmatrix} \frac{\cos(\theta-\alpha)}{\cos\alpha} & \rho\sin(\theta) \\ -\frac{\sin(\theta-2\alpha)}{\rho\cos^2(\alpha)} & \frac{\cos(\theta-\alpha)}{\cos\alpha} \end{bmatrix}$$

Hint:

$$\cos(\theta) + \tan(\alpha)\sin(\theta) = \frac{\cos(\theta-\alpha)}{\cos(\alpha)}$$
$$\sin(\theta) - \tan(\alpha)\cos(\theta) = \frac{\sin(\theta-\alpha)}{\cos(\alpha)}$$

SOLUTION: Dipole Edge Corrections - Horizontal20 pts.

a) 15 pts: Start by defining the geometry of the wedge:



$$\ell \cos\left(\frac{\alpha}{2}\right) = x$$
$$\ell \sin\left(\frac{\alpha}{2}\right) = \frac{s}{2}$$
$$s = 2\ell \sin\left(\frac{\alpha}{2}\right) = 2x \tan\left(\frac{\alpha}{2}\right) \simeq x \tan(\alpha)$$

Assume γ, $v \simeq=$ const, and calculate the impulse across the wedge.

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = q\mathbf{v} \times \mathbf{B}$$
$$m\gamma v^2 \frac{\mathrm{d}^2 x}{\mathrm{d}s^2} \simeq q(\mathbf{v} \times \mathbf{B}) \cdot \hat{\mathbf{x}} = -qvB_y$$
$$\implies m\gamma v^2 \int_{\text{wedge}} \frac{\mathrm{d}}{\mathrm{d}s}\frac{\mathrm{d}x}{\mathrm{d}s}\, ds = -qv \int_{\text{wedge}} B_y\, ds$$

where we have used

$$\frac{d^2x}{dt^2} = \frac{d^2s}{dt^2}\frac{dx}{ds} + \left(\frac{ds}{dt}\right)^2 \frac{d^2x}{ds^2}, \qquad \frac{d^2s}{dt^2} = 0, \qquad \frac{ds}{dt} = v$$

to change the independent variable.

Evaluate the 1st integral and approximate the second integral above:

$$\int_{\text{wedge}} \frac{\mathrm{d}}{\mathrm{d}s}\frac{\mathrm{d}x}{\mathrm{d}s}\, ds = \int_{\text{wedge}} \frac{\mathrm{d}}{\mathrm{d}s} x'\, ds = \Delta x' = \text{Change in } x\text{-angle}$$
$$B_y|_{\text{wedge}} = -B_y(0) \implies \int_{\text{wedge}} B_y\, ds \quad \simeq -B_y(0) \cdot s$$

Therefore:

$$\Delta x' = \frac{qB_y(0)}{m\gamma v} \tan(\alpha) x$$
$$= \frac{B_y(0)}{[B\rho]} \tan(\alpha) x$$

with $[B\rho] = p/q = m\gamma v/q$ denoting the rigidity. Integrating again:

$$\Delta x = 0$$

This result can be expressed in terms of a transfer matrix:

$$\mathbf{M}_{\text{wedge}} = \begin{bmatrix} 1 & 0 \\ \frac{\tan(\alpha)}{\rho} & 1 \end{bmatrix}$$

since the bend radius ρ satisfies $\frac{1}{\rho} = \frac{B_y(0)}{[B\rho]}$.

b) 5 pts:

$$\begin{aligned}
\mathbf{M}_x &= \mathbf{M}_{\text{wedge}} \cdot \mathbf{M}_{\text{sector}} \cdot \mathbf{M}_{\text{wedge}} \\
&= \begin{bmatrix} 1 & 0 \\ \frac{\tan(\alpha)}{\rho} & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\theta) & \rho\sin(\theta) \\ -\frac{\sin(\theta)}{\rho} & \cos(\theta) \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \frac{\tan(\alpha)}{\rho} & 1 \end{bmatrix} \\
&= \begin{bmatrix} \frac{\cos(\theta-\alpha)}{\cos(\alpha)} & \rho\sin(\theta) \\ \frac{\sin(\alpha)\cos(\theta-\alpha)}{\rho\cos^2(\alpha)} - \frac{\cos(\alpha)\sin(\theta-\alpha)}{\rho\cos^2(\alpha)} & \tan(\alpha)\sin(\theta) + \cos(\theta) \end{bmatrix} \\
&= \begin{bmatrix} \frac{\cos(\theta-\alpha)}{\cos(\alpha)} & \rho\sin(\theta) \\ -\frac{\sin(\theta-2\alpha)}{\rho\cos^2(\alpha)} & \frac{\cos(\theta-\alpha)}{\cos(\alpha)} \end{bmatrix}
\end{aligned}$$

Here we applied the trigonometric identity:

$$\begin{aligned}
\sin(\theta - 2\alpha) &= \sin(\theta - \alpha - \alpha) \\
&= \sin(\theta - \alpha)\cos(\alpha) - \cos(\theta - \alpha)\sin(\alpha)
\end{aligned}$$

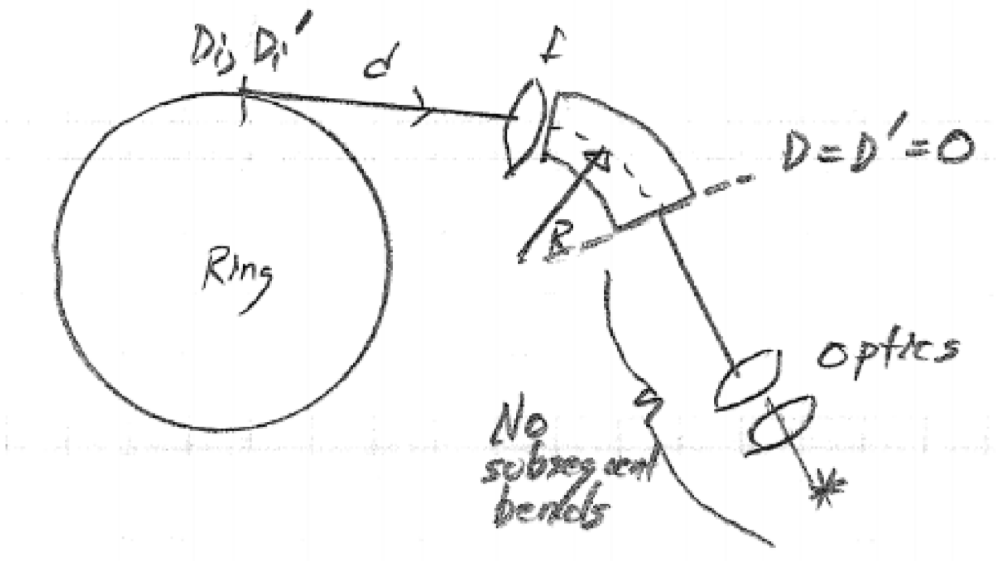
Problem 3. Dispersion In Extraction Line 20 pts.

The dispersion function in a period ring satisfies:

$$D''(s) + K(s)D(s) = \frac{1}{\rho(s)}$$
$$\rho(s) = \text{ bend radius}, K(s) = \text{ focusing function}$$
$$D\left(s + L_p\right) = D(s), \rho\left(s + L_p\right) = \rho(s), K\left(s + L_p\right) = K(s)$$
$$L_p = \text{ lattice period}$$

a) 5 pts: Argue that the solution for D is unique. This implies that there is a unique closed orbit $x = \delta D$ for every value of off-momentum δ. (Hint: let D_1 andf D_2 be two independent solutions and look for a contradiction.)

b) 15 pts: A particle is kicked out of a ring with dispersion $D = D_i$ and $D' = D_i'$ just after the kick. The particle is then transported through an extraction line with a drift length d, a thin lens focusing kick with focal length f, and a sector bend of radius $\rho = R$ and length ℓ, and finally through an unspecified series of optics to the target. (See diagram below).



Find a constraint on the lattice parameters d, f, R, and ℓ that can be enforced to ensure that $D = D' = 0$ after the bending magnet (as indicated) to have zero dispersion in the straight transport and focusing line to the target. (Assume the elements can all be treated in the thin approximation.)

SOLUTION: Dispersion In Extraction Line 20 pts.

a) 5 pts: Let $D_1(s)$ and $D_2(s)$ be independent, non-zero solutions for the dispersion. Then any linear combination of D_1 and D_2 is also a solution. Let $D(s) = D_2(s) - D_1(s)$.

$$D''(s) + K(s)D(s) = \left(D_2''(s) - D_1''(s)\right) + K(s)\left(D_2(s) - D_1(s)\right) = \frac{1}{\rho(s)} - \frac{1}{\rho(s)} = 0$$

This indicates there is no dispersion, which is a contradiction to our assumption. Therefore, the solution must be unique.

b) 15 pts: The 3×3 transfer matrix for the drift, focus, and bend is:

$$\mathbf{M} = \mathbf{M}_B\mathbf{M}_F\mathbf{M}_D = \begin{bmatrix} 1-\frac{\ell}{f} & d+\ell\left(1-\frac{d}{f}\right) & \frac{\ell^2}{2R} \\ -\frac{1}{f} & 1-\frac{d}{f} & \frac{\ell}{R} \\ 0 & 0 & 1 \end{bmatrix}$$

To suppress the dispersion at the end of this section, we require:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \mathbf{M}\begin{bmatrix} D \\ D' \\ 1 \end{bmatrix}$$

$$\left(1-\frac{\ell}{f}\right)D_i + \left[d+\ell\left(1-\frac{d}{f}\right)\right]D_i' + \frac{\ell^2}{2R} = 0$$

$$\boxed{-\frac{D_i}{f} + \left(1-\frac{d}{f}\right)D_i' + \frac{\ell}{R} = 0}$$